

Project 1

Differentiated Demand for Cars

AMN & NV

Due: Wednesday, October 21st at 17:00

1 Empirical Part

1.1 Data

The dataset, `cars.csv`, covers five European countries (Belgium, France, Germany, Italy and the UK) over the thirty years 1970–1999. Throughout, we refer to a country–year pair as a *market*, indexed by t , so there are 150 markets. From the raw underlying dataset, the $J = 40$ highest-selling cars have been selected in every market; that is, there are always precisely 40 inside alternatives available in any market t , although *which* 40 cars they are changes across countries and years. There are therefore $150 \times 40 = 6,000$ rows in `cars.csv`, or 1,200 rows per country. The data are a subset of those used by [Goldberg and Verboven \(2001\)](#).

Appendix A presents a full list of variables with labels and the mean values (for numeric variables). Perhaps the most important variables to consider include the price (`pr`), car weight (`we`), fuel efficiency (`li`), quantity sold (`qu`), the country’s population (`pop`), the segment (`cla`), and the domestic-car dummy (`home`).

1.2 Questions

The overarching goal of this project is to imagine that you are working for the Competition Authority, having just received a formal application from VW to merge with Fiat. You must decide whether to permit this merger. Thus, the research question is:

- **Overall question:** *What would be the effects on prices and consumer welfare from a merger between VW and Fiat?*

To answer this question, please proceed along the following lines:

- **Question 1 (demand):** Estimate a simple logit and an extended logit model of the consumer demand for cars. Report parameter estimates, but also give the reader a sense of the own- and cross-price elasticities implied by each model, and of the diversion ratios between VW’s and Fiat’s products.

Here, “extended” means a model that improves on the simple logit in a way you can name. There are two broad directions, and you may combine them:

- (a) *A more credible price coefficient:* fixed effects and/or instrumental variables, still within the simple logit.
- (b) *Richer substitution patterns:* a nested logit or a random coefficients logit (with or without instruments).

You do not have to address every concern covered in the course (endogeneity, heterogeneity, etc.). You do have to say which concern your extension addresses, why you think it is the one that matters for this merger, and whether the data bear that out.

- **Question 2 (merger):** Conduct the merger counterfactual for both of your models, and analyse similarities and differences. Compare the effects of the merger across markets. How large would cost savings have to be to make the merger net positive for consumers? Where the two models disagree, trace the disagreement back to the demand side: is it driven by what the models *estimate* differently (e.g. the price coefficient), or by what one model *rules out* that the other allows (e.g. substitution patterns beyond IIA)?

In setting up your model, you will have to make a number of practical decisions. The notes below are pointers, not instructions; there are many defensible specifications, and some indefensible ones.

- **Market size:** The variable *qu* gives the quantity of sold units, and *pop* is the total population of the country (*ma*) in the given year (*ye*). However, there are several ways of defining a market share depending on what one views as the “market size”, M_t . Is the market the full population? Just the driving-age population? Individuals or households?

Hint: Shares must leave room for an outside option, so dividing qu by total sales in the market will not do. See Section 1 of the course note (What a demand system has to do) for why M_t matters for your merger results.

- **Price variable:** We recommend using the log of the price, but you could also use the price in levels, or normalised by income (*princ*) or converted to a common currency (*eurpr*).

Hint: pr is measured in nominal local currency, including VAT. Make sure the price variation your regression actually uses is variation you would be happy to explain. If price enters in logs, converting a change in consumer surplus into money takes a little care (course note, Section 3, Supply and Equilibrium).

- **Controls and fixed effects:** What variables to include in x_{jt} . We recommend including the car’s weight (*we*), fuel efficiency (*li*), and fixed effects for the brand (*brd*). But there are many others you can choose from, and the set of fixed effects is a choice in its own right.

Hint: The choice of fixed effects is likely where your price coefficient moves the most, including its sign. Report which fixed effects you include and what price variation they leave you with (course note, Section 4, Where Prices Come From: Endogeneity and Instruments).

- **Instrument(s):** If you choose to use instrumental variables, there are variables relating to the exchange rates, as well as the VAT rate (*tax*). You can also construct the “differentiation IVs” (à la [Berry et al., 1995](#) or [Gandhi and Houde, 2019](#)).

Hint: For each candidate, ask in which dimension of the panel it varies (car, country, year) and whether your fixed effects absorb that dimension. Report the first-stage F-statistic. A nested logit needs a second excluded instrument for the within-nest share (course note, Section 5, Richer Demand I: Nested Logit); a random coefficients logit needs over-identifying instruments that move a car’s competitive environment (course note, Section 7, Estimation: One Algorithm). Count your moments against your parameters.

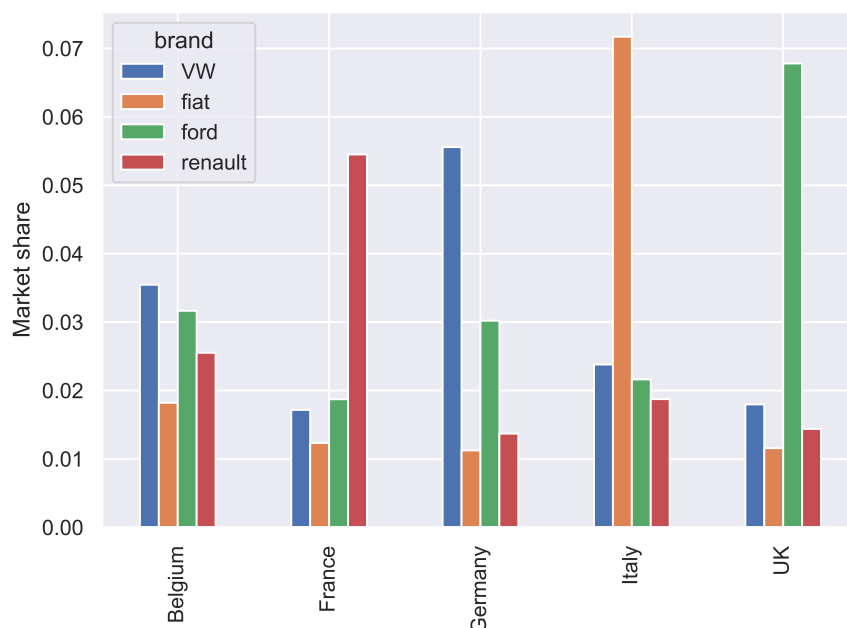
- **Substitution patterns:** If you estimate a nested logit, you must choose the nests; if you estimate a random coefficients logit, you must choose which coefficients are random.

*Hint: Candidate nests include the segment (*cla*) and domestic vs. foreign. Which coefficient you make random partly determines your markups before you have seen a single estimate (course note, Appendix D, Properties of Logit Demand). You may use *pyblp* or your own code from PS3 and PS4. Data on the income distribution by country and year are also available, should you want tastes to vary with income; using them is by no means a requirement.*

You must also include at least one robustness check: pick the one decision above that you are least sure of and show how the merger results change under a plausible alternative. Remember to argue (briefly) for your choices!

As inspiration for your analysis, you may for example choose to emphasise some notion of a “home market bias.” Figure 1 shows a simple descriptive graph that motivates why we might suspect home bias to be strong: the figure shows the market shares of four selected car manufacturers in each of the five countries in the dataset. A clear pattern emerges: each manufacturer has by far the strongest sales in its home market. Belgium serves as a neutral comparison since that country has no domestic car manufacturing. In the simplest model, home bias is just higher demand for domestic cars; but domestic buyers may also differ in how *price sensitive* they are, or in how readily they substitute between domestic and foreign cars. Each of these has different implications for where the VW–Fiat merger does its damage.

Figure 1: Home Bias and Car Market Shares



2 Theoretical Questions

- 2.1 Consider the theoretical models of price competition with homogeneous products seen so far in the course. Can any of these models (possibly after a slight adjustment) potentially rationalize the observed market shares from the data for this project? If so, which model(s)? Explain and justify your answer verbally, i.e. you are not expected here to run any regressions.
- 2.2 Consider the Perloff–Salop model of price competition with horizontally differentiated products, as seen in Lecture 1 and Problem Set 1. Summarize what is/are the assumed source(s) of product differentiation in this theoretical model. Then describe the assumed sources of product differentiation in the empirical models that you implemented for this project, commenting on any similarities or differences.
- 2.3 Test empirically whether you can reject the Perloff–Salop model (perhaps with slight adjustments regarding e.g. error terms, compared to the presentation in Lecture 1) using the data for this project.

Hint: Formulate a joint null hypothesis and explain why rejecting this hypothesis can be interpreted as rejecting the Perloff–Salop model. It is sufficient to conduct the joint test using a standard linear regression.

- 2.4 Now consider a counterfactual situation where the number of firms selling cars is very large. Based on insights from the Perloff–Salop model, what do you expect would happen to prices in this situation? Would your answer differ depending on whether ‘taste shocks’ were drawn from a normal distribution (Gaussian), instead of a Type I Extreme Value distribution (Gumbel)? If so, comment on how this difference might affect the way you interpret the coefficients obtained from the kind of empirical discrete choice analysis carried out for this project.

The oral exam

This project may be drawn for your oral exam. If it is, then the questions will concern both the empirical and the theoretical parts of the project. For the empirics, the examination will start from *your* choices: the market size, the price variable, the fixed effects, the instruments, the nests or random coefficients, and the conduct assumption behind the merger simulation. For each, expect to be asked what you assumed, why it is reasonable here, and what would happen to the merger results if it were false. A simple model that you can defend is worth more than an advanced one that you cannot. For the theory, the examination will start from *your* arguments for how and why the relevant theoretical models relate to the empirical exercise, and your choices regarding testing.

3 Formal Requirements

- **Format:** The report must be written in English and uploaded to FeedbackFruits via Absalon as a single PDF file.
- **Size:** Maximum 7 pages + 2 pages of output.
 - A page has fontsize = 12pt, line spacing = 1.5, and page margins 2.5 cm. Output includes tables and figures and must be formatted and labelled appropriately. I.e. make sure to choose an appropriate number of decimals and to use sensible variable labels.
 - *Note:* There is no requirement to hit the maximum number of pages. For example, 6 pages may be sufficient. Use your judgement.
- **Feedback:** After the submission deadline, you are expected to read at least one other student/group’s project and provide them with written feedback via FeedbackFruits. Please comment both on what is done well and what can be clarified or improved. Be constructive!
 - **Deadline:** The deadline for providing this feedback is **Wednesday October 28 at 17:00**, exactly one week after the submission deadline. Providing this feedback is a formal requirement to qualify for the final exam.
- **Code:** You must submit your Python code alongside your report. Make sure that your code runs without modifications as far as possible.¹ Every number in the report should be reproducible from it.
- **Groups:** You are allowed (and strongly encouraged) to work in groups of up to three people. List all group members on the front page of your report in alphabetical order

¹For instance, make sure to use relative paths (i.e. not C:\user\long\path, but ./input/cars.csv).

of surnames. The oral exam is individual: every group member must be able to defend every choice in the report.

- **AI tools:** Use of AI tools is allowed when working on and writing up the project. If you use AI tools, then you must include a brief explanation in the project as to how these tools were used.² We do not recommend directly including text in the project that was generated (solely or mainly) by an AI tool. However, if you choose to include such text, then the tool must be quoted as a source. AI tools may help you implement a more advanced model than you could have coded yourself; that is only appropriate if you can explain and justify that model at the oral exam.

References

- Steven Berry, James Levinsohn, and Ariel Pakes. Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890, 1995.
- Amit Gandhi and Jean-François Houde. Measuring substitution patterns in differentiated-products industries. NBER Working Paper 26375, National Bureau of Economic Research, 2019. Revised 2020.
- Pinelopi Koujianou Goldberg and Frank Verboven. The evolution of price dispersion in the european car market. *Review of Economic Studies*, 68(4):811–848, 2001.

²For example: “I have used Github copilot to get suggestions for code, and I have made ChatGPT write first drafts for paragraphs for text based on my prompt, which I have used as a starting point for writing paragraphs of my own.”

A Variable Labels

Variable	Label	Mean
ye	year (=first dimension of panel)	84.50
ma	market (=second dimension of panel)	3.00
co	model code (=third dimension of panel)	207.50
zcode	alternative model code (predecessors and successors get the same number)	177.76
brd	brand code	16.79
type	name of brand and model	
brand	name of brand	
model	name of model	
org	origin code (demand side: country with which consumers associate the model)	2.72
loc	location code (production side: country where the model is produced)	5.17
cla	class or segment code	2.30
home	domestic car dummy (appropriate interaction of org and ma)	0.32
frm	firm code	14.50
qu	sales (number of new car registrations)	35606.68
cy	cylinder volume or displacement (in cc)	1337.09
hp	engine power (in kW)	50.10
we	weight (in kg)	934.49
pl	places (number, not reliable variable)	4.88
do	doors (number, not reliable variable)	3.55
le	length (in cm)	409.24
wi	width (in cm)	163.44
he	height (in cm)	140.46
li1	fuel consumption, measure 1 (litres per 100 km, at 90 km/h)	6.59
li2	fuel consumption, measure 2 (litres per 100 km, at 120 km/h)	8.11
li3	fuel consumption, measure 3 (litres per 100 km, at city speed)	8.92
li	average of li1, li2, li3 (used in papers)	7.87
sp	maximum speed (km/hour)	154.22
ac	time to acceleration (in seconds from 0 to 100 km/h, some from 0 to 96 km/h)	16.27
pr	price (in destination currency, including VAT)	2608988.58
princ	=pr/(ngdp/pop): price relative to per capita income (often used in demand models)	0.76
eurpr	=pr/avdextr: price in common currency (in SDR; multiply by 1.2956 to interpret in euros)	7256.92
exppr	=pr/avexr: price in exporter currency	600384.65
avexr	av. exchange rate of exporter country (exporter loc currency per SDR)	229.30
avdextr	av. exchange rate of destination country (destination ma currency per SDR)	319.91
avcpr	av. consumer price index of exporter country	492.53
avppr	av. producer price index of exporter country	671.77
avdcpr	av. consumer price index of destination country	77.02
avdppr	av. producer price index of destination country	87.69
xexr	=avdextr/avexr	66.32
tax	VAT rate (as a fraction, e.g. 0.21 = 21%)	0.21
pop	population	49183800.00
ngdp	nominal gross domestic product of destination country (destination currency)	178667304825541.97
rgdp	real gross domestic product	216716720230172.44
engdp	=ngdp/avdextr: nominal GDP in common currency (SDR)	504371708122.45
ergdp	=rgdp/avexr	659002565347.56
engdpc	=engdp/pop: nominal GDP per capita in common currency	10015.84
ergdpc	=ergdp/pop	13264.84
inc	avg. income per capita	26829.74